

Invariant Information Clustering for Unsupervised Image Classification and Segmentation

Department of Computer Engineering and Artificial Intelligence
MAI Lab.

YS Park



ICCV 2019 open access

These ICCV 2019 papers are the Open Access versions, provided by the [Computer Vision Foundation](#).

Except for the watermark, they are identical to the accepted versions; the final published version of the proceedings is available on IEEE Xplore.

This material is presented to ensure timely dissemination of scholarly and technical work. Copyright and all rights therein are retained by authors or by other copyright holders. All persons copying this information are expected to adhere to the terms and constraints invoked by each author's copyright.

Invariant Information Clustering for Unsupervised Image Classification and Segmentation

Xu Ji, Joao F. Henriques, Andrea Vedaldi; Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019, pp. 9865-9874

Abstract

We present a novel clustering objective that learns a neural network classifier from scratch, given only unlabelled data samples. The model discovers clusters that accurately match semantic classes, achieving state-of-the-art results in eight unsupervised clustering benchmarks spanning image classification and segmentation. These include STL10, an unsupervised variant of ImageNet, and CIFAR10, where we significantly beat the accuracy of our closest competitors by 6.6 and 9.5 absolute percentage points respectively. The method is not specialised to computer vision and operates on any paired dataset samples; in our experiments we use random transforms to obtain a pair from each image. The trained network directly outputs semantic labels, rather than high dimensional representations that need external processing to be usable for semantic clustering. The objective is simply to maximise mutual information between the class assignments of each pair. It is easy to implement and rigorously grounded in information theory, meaning we effortlessly avoid degenerate solutions that other clustering methods are susceptible to. In addition to the fully unsupervised mode, we also test two semi-supervised settings. The first achieves 88.8% accuracy on STL10 classification, setting a new global state-of-the-art over all existing methods (whether supervised, semi-supervised or unsupervised). The second shows robustness to 90% reductions in label coverage, of relevance to applications that wish to make use of small amounts of labels. github.com/xu-ji/IIC

- Introduction
- Related Work
- Method
- Experiments
- Conclusion

- Most Supervised DL methods require large quantities of **manually labelled data**.
 - This is true for large-scale image classification and even more for segmentation
 - the annotation cost per image is very high.
- Unsupervised clustering aims to group data points into classes entirely **without labels**.
 - K-means + DL(representation) is main approach.
 - The main issue of this is **degenerate solutions**(empty cluster).
 - To address this involving pre-training, feature post-processing (whitening or PCA), clustering mechanisms external to the network.
 - This paper introduces **Invariant Information Clustering (IIC)**, a method that addresses this issue in a more principled manner.

- Simple objective function
 - **Mutual Information(MI)** between the function's classifications for paired data samples.
 - Since the clustering space is discrete, mutual information can be computed exactly.
- IIC is intrinsically robust to **two issues** that affect other methods.
 - Clustering Degeneracy
 - Due to the **entropy maximization**
 - the loss is not minimised if all images are assigned to the same class
 - Noisy data with unknown or distractor classes
 - IIC addresses this issue by employing an auxiliary output overclustering layer (multi-head).

- **Co-clustering and mutual information**
 - The use of information as a criterion to learn representations is **not new**.
 - IMSAT, Deep InfoMax: high computation(continuous variables), complex loss function.
 - IIC : simple and exact computations(discrete variables), only MI.
- **Semantic clustering vs intermediate representation learning**
 - Semantic clustering generates **directly** outputs (semantic) clusters.
 - intermediate representation learning generates feature that needs post-processing.
- **Optimising image-to-image distance**
 - Many approaches to deep clustering utilise a distance function between input images that approximates a given grouping criterion.
 - k-means style mechanisms need explicit prevention degenerate phenomenon
 - pre-training, cluster-reassignment or feature cleaning via PCA and whitening.
- **Invariance as a training objective**
 - Treat X and $g(X)$ as equivalent $\rightarrow$ network learns consistent representations

- The goal of IIC is to learn a representation $\Phi: X \rightarrow Y$
 - $Y = \{1, \dots, C\}$
 - $x, x' \in X$
 - $$I(z, z') = \sum_{c=1}^C \sum_{c'=1}^C P_{cc'} \ln \frac{P_{cc'}}{P_c P_{c'}} \\ = H(z) - H(z | z')$$
 - $\max_{\phi} I(\phi(x), \phi(x'))$

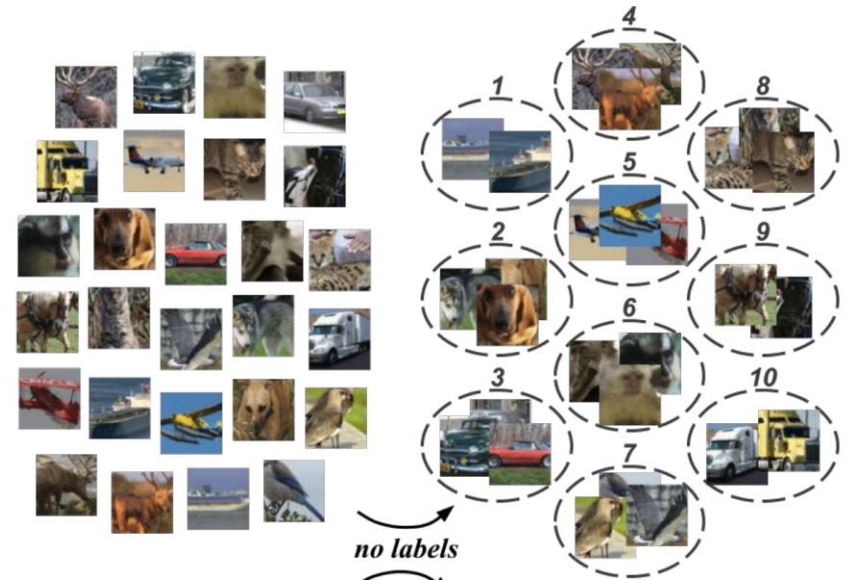
- The goal of IIC is to learn a representation $\Phi: X \rightarrow Y$

- $Y = \{1, \dots, C\}$

- $x, x' \in X$

- $$I(z, z') = \sum_{c=1}^C \sum_{c'=1}^C P_{cc'} \ln \frac{P_{cc'}}{P_c P_{c'}}$$
$$= H(z) \uparrow - H(z | z') \downarrow$$

- $\max_{\phi} I(\phi(x), \phi(x'))$
 - Prevent degenerate solution
 - Prediction consistency
 - Maintain shared content
 - Remove instance-specific
 - How about without one of them?



- $\max_{\phi} I(\phi(x), \phi(gx))$
 - scaling, skewing, rotation or flipping (geometric)
 - changing contrast and colour saturation (photometric)
 - To learn **invariant** factor.
- Auxiliary overclustering
 - contain only relevant
 - contain irrelevant or distractor classes

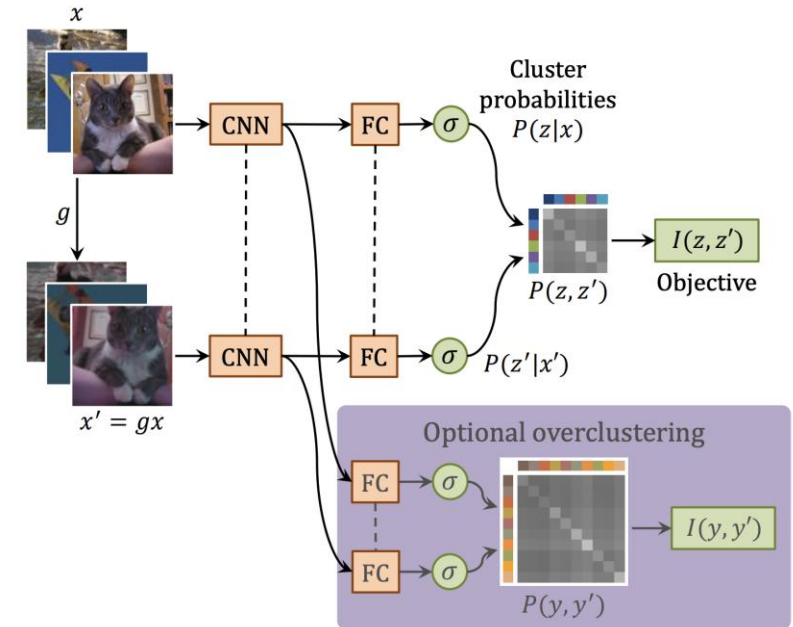


Figure 2: IIC for image clustering. Dashed line denotes shared parameters, g is a random transformation, and I denotes mutual information (eq. (3)).

- two modifications.
 - clustering is applied to image patches
 - one has access to the spatial relationships between patches.

$$\max_{\Phi} \frac{1}{|T|} \sum_{t \in T} I(P_t),$$

$$P_t = \frac{1}{n |G| |\Omega|} \sum_{i=1}^n \sum_{g \in G} \sum_{u \in \Omega} \Phi_u(x_i) \cdot ([g^{-1} \Phi(gx_i)]_{u+t})^\top$$

- Φ : CNN output, patch-wise cluster probability vector
- x_i : the i -th image
- G : set of transformations (geometric/photometric)
- Ω : set of all pixel locations in the image
- T : set of neighbor patch displacements
- P_t : joint probability matrix for displacement t
- $I(P_t)$: mutual information of the joint probability matrix P_t

- Datasets
 - STL10, CIFAR10, CIFAR100-20, MNIST
 - Fully unsupervised: train on full dataset, test on labelled subset
 - Data augmentation: random color t.f. + horizontal flips
 - batch repetition $r \in [1,5]$ for more pairs
- Architecture
 - Base CNN: ResNet or VGG11-like
 - Output heads: main head ($k=k_{gt}$) + auxiliary overclustering head ($k>k_{gt}$)
 - Each head duplicated $h = 5$ sub-heads
 - Outputs: probability distribution over clusters for each image
- Training
 - Optimizer: Adam, $lr = 1e-4$
 - IIC loss maximization (eq. 3)
 - Alternate training main/aux heads per epoch

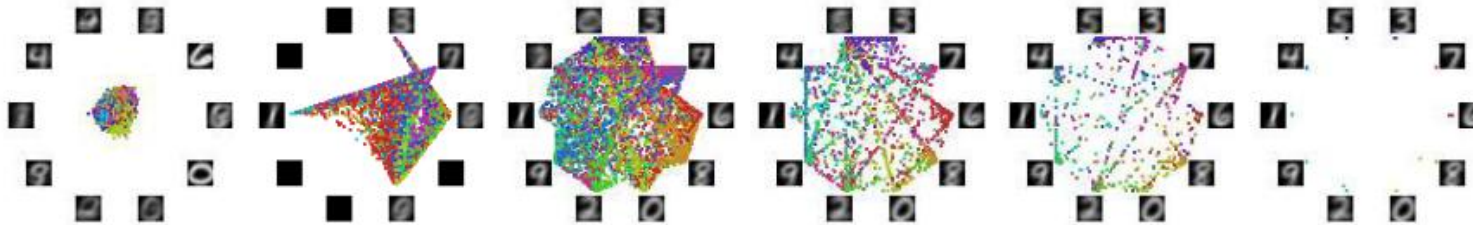


Figure 3: Training with IIC on unlabelled MNIST in successive epochs from random initialisation (left). The network directly outputs cluster assignment probabilities for input images, and each is rendered as a coordinate by convex combination of 10 cluster vertices. There is no cherry-picking as the entire dataset is shown in every snapshot. Ground truth labelling (unseen by model) is given by colour. At each cluster the average image of its assignees is shown. With neither labels nor heuristics, the clusters discovered by IIC correspond perfectly to unique digits, with one-hot certain prediction (right).

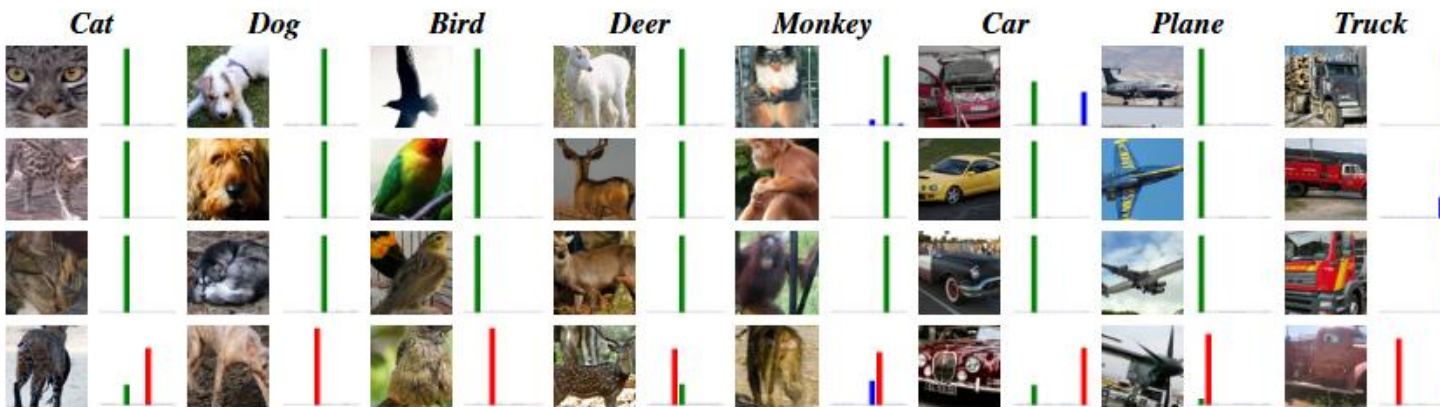


Figure 5: Unsupervised image clustering (IIC) results on STL10. Predicted cluster probabilities from the best performing head are shown as bars. Prediction corresponds to tallest, ground truth is green, incorrectly predicted classes are red, and all others are blue. The bottom row shows failure cases.

	STL10	CIFAR10	CFR100-20	MNIST
Random network	13.5	13.1	5.93	26.1
K-means [53]†	19.2	22.9	13.0	57.2
Spectral clustering [49]	15.9	24.7	13.6	69.6
Triplets [46]‡	24.4	20.5	9.94	52.5
AE [5]‡	30.3	31.4	16.5	81.2
Sparse AE [40]‡	32.0	29.7	15.7	82.7
Denoising AE [48]‡	30.2	29.7	15.1	83.2
Variational Bayes AE [34]‡	28.2	29.1	15.2	83.2
SWWAE 2015 [54]‡	27.0	28.4	14.7	82.5
GAN 2015 [45]‡	29.8	31.5	15.1	82.8
JULE 2016 [52]	27.7	27.2	13.7	96.4
DEC 2016 [51]†	35.9	30.1	18.5	84.3
DAC 2017 [8]	47.0	52.2	23.8	97.8
DeepCluster 2018 [7]† ‡	33.4★	37.4★	18.9★	65.6★
ADC 2018 [24]	53.0	32.5	16.0★	99.2
IIC (lowest loss sub-head)	59.6	61.7	25.7	99.2
IIC (avg sub-head $\pm$ STD)	59.8 ± 0.844	57.6 ± 5.01	25.5 ± 0.462	98.4 ± 0.652

Table 1: Unsupervised image clustering. Legend: †Method based on k-means. ‡Method that does not directly learn a clustering function and requires further application of k-means to be used for image clustering. ★Results obtained using our experiments with authors’ original code.

	STL10
No auxiliary overclustering	43.8
Single sub-head ($h = 1$)	57.6
No sample repeats ($r = 1$)	47.0
Unlabelled data segment ignored	49.9
Full setting	59.6

Table 2: Ablations of IIC (unsupervised setting). Each row shows a single change from the full setting. The full setting has auxiliary overclustering, 5 initialisation heads, 5 sample repeats, and uses the unlabelled data subset of STL10.

- Datasets
 - COCO-Stuff-3 (sky, ground, plants) & COCO-Stuff (15 coarse labels)
 - Potsdam-3 (sky/ground/vegetation) & Potsdam-6 (6 labels)
 - Input images: cropped & resized (128×128 for COCO, 200×200 for Potsdam)
 - Sobel preprocessing & ignore non-stuff pixels
- Architecture
 - Base CNN + 1×1 conv layer heads
 - Overclustering: $k = 3 \sim 5 \times k_{gt}$
 - Single sub-head (h=1), batch repetition r=1
- Training
 - Convolutional l1C loss, displacement d=10
 - Optional entropy coefficient = 1.5 for Potsdam/COCO-Stuff-3
 - Optimisation over all pixels in batch simultaneously



Figure 7: Example segmentation results (un- and semi-supervised). Left: COCO-Stuff-3 (non-stuff pixels in black), right: Potsdam-3. Input images, IIC (fully unsupervised segmentation) and IIC* (semi-supervised overclustering) results are shown, together with the ground truth segmentation (GT).

	COCO-Stuff-3	COCO-Stuff	Potsdam-3	Potsdam
Random CNN	37.3	19.4	38.2	28.3
K-means [44]†	52.2	14.1	45.7	35.3
SIFT [39]‡	38.1	20.2	38.2	28.5
Doersch 2015 [17]‡	47.5	23.1	49.6	37.2
Isola 2016 [30]‡	54.0	24.3	63.9	44.9
DeepCluster 2018 [7]† ‡	41.6	19.9	41.7	29.2
IIC	72.3	27.7	65.1	45.4

Table 4: **Unsupervised segmentation.** IIC experiments use a single sub-head. Legend: †Method based on k-means. ‡Method that does not directly learn a clustering function and requires further application of k-means to be used for image clustering.



Figure 3: Additional unsupervised segmentation (IIC) results on COCO-Stuff-3 (non-stuff pixels in black). Left to right for each triplet: image, prediction, ground truth.

- Key Idea:
 - Train model without labels or heuristics.
 - Learn semantic clusters directly from data.
- Core Objective:
 - Maximise MI.
 - Related pairs generated via random transformations.
 - Avoids degenerate solutions naturally.
- Results:
 - Achieves SOTA(6 years ago ...) semantic accuracy in clustering and segmentation.
 - Fully unsupervised.
- Broader Implications:
 - Not limited to vision.
 - Can be applied to temporal data streams and general representation learning.